Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/24/2019
PI Name:	Stenger, Michael Ph.D.		
Project Title:	Venous Congestion Countermeasure Study		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	NOTE Aug 2018: Previously with KBRwyle at Johnson Space Center		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/18/2018	**End Date:**	02/16/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/16/2020 due to PI change from M. Stenger to K. Marshall-Goebel (Ed., 7/10/2020)		
Key Personnel Changes/Previous PI:	July 2020: Principal Investigator (PI) Dr. Michael Stenger moved to Research Operations and Integration element in February 2020; project taken over at that time by Dr. Karina Marshall-Goebel. May 2019 report: Add Karina Marshall-Goebel, Ph.D., as Co-investigator.		
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Grant/Contract No.:	Directed Research		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The chronic headward fluid shift induced by weightlessness is hypothesized to be the instigating factor for the development of ocular structural and functional changes that develop during long-duration spaceflight. Efforts to determine a possible countermeasure (CM) in ground-based studies have focused on reversing the headward fluid shift using a variety of mechanical approaches, including lower body negative pressure (LBNP), veno-constrictive thigh cuffs (VTC), and reducing intrathoracic pressure during inspiration using an impedence threshold device (ITD). However, a final CM has not been implemented during spaceflight, in part because it is unknown which single or combination of CMs will be most efficacious at reversing the headward fluid shift. Measurements of internal jugular vein cross sectional area and pressure (IJVA and IJVP, respectively) collected on Earth in subjects in the 15° head-down tilt (HDT) position indicate that LBNP, VTC, or ITD partially reverse the headward fluid shift to values similar to the supine position, but no single CM successfully reversed the headward fluid shift during HDT to levels observed during upright posture. Moreover, it is critical to determine how best to provide sustained relief of the headward fluid shift. The purpose of this project is to determine if individual CMs or a combination of CMs can effectively reduce the headward fluid shift induced by supine posture and to determine the effectiveness of prolonged use of a CM during sleep. Specific Aims: 1. Determine if a single or combination of mechanical countermeasures, including LBNP, VTC, and breathing through an ITD, will acutely reverse a seated-to-supine posture headward fluid shift and sustain it for up to 30 min. Outcomes will include noninvasive measures of ICP, IJVA, IJVP, choroid thickness, and intraocular pressure. 2. Determine if the most effective combination of CMs to reverse vascular outcomes in SA#1 also reverses the cerebral spinal fluid (CSF) shift by quantifying intraventricular volumes using magnetic resonance imaging (MRI). 3. Determine if up to 8 hours of daily LBNP during sleep can prevent the headward fluid shift-induced changes during 3 days of head-down tilt bedrest. Research Methods: Direct intracranial pressure (ICP) measured during parabolic flight [Lawley et al., J Physiol, 595:2115-27, 2017] and preliminary noninvasive measures of ICP collected during long-duration spaceflight (Fluid Shifts Study) suggest ICP during weightlessness is similar to or below the supine position in a 1-g environment; IJVA and IJVP measured during spaceflight are also similar to values measured in the supine posture on Earth. These data suggest use of the supine posture on Earth is a more appropriate posture for CM evaluation so as to not confound their effectiveness due to the hydrostatic column which still exists in ground-based studies. To accomplish Aim #1 we will evaluate the effectiveness of candidate CMs to reverse the headward fluid shift imposed by moving from the seated to the supine posture. Ten subjects will participate in 3 visits to the NASA Johnson Space Center (JSC) Cardiovascular and Vision Laboratory. Each day subjects will be studied in the seated and supine position (baseline) and in the supine position during and after the application of each CM separately (Day 1: LBNP, TC, ITD, randomized order) or CM combinations (Days 2 and 3: LBNP+TC, LBNP+ITD, VTC+ITD, and LBNP+ITD+VTC.). Each data collection phase is depicted by a series of blocks that indicate the condition. Each phase begins with 5 minutes of stabilization, followed by ocular and vascular ultrasound imaging, IJVP, noninvasive ICP measurements, bilateral IOP measurement, and OCT imaging. All measurements are repeated during each phase. After application of single CMs, measurements will continue during the 30-min post-CM phase. For combination CMs, the primary CM (LBNP or ITD) is applied for 30 min followed by the secondary CM (VTC, ITD, or VTC+ITD). This will allow us to determine if sequestering fluid in the lower body using an “active” CM, like LBNP, can be maintained using more “passive” methods, such as VTC or ITD. Operationally these “passive” CMs have the advantage of allowing crewmembers to move freely throughout the vehicle and perform other tasks without being restrained by a CM device. Thus, we will determine how best to sustain the effects of the acute application of CMs. To accomplish Aim 2, co-investigators from the University of California San Diego and UT (University of Texas) Health in Houston will use magnetic resonance imaging (MRI) to determine the impact of countermeasures employed in Aim 1 have on cerebral spinal fluid volume. Subjects will complete testing in the supine position with and without CMs, and total CSF volume will be quantified. To accomplish Aim 3, co-investigators at UT Southwestern will develop a LBNP device that can be worn for 8 hours during sleep and test this CM throughout a 3-day bedrest study. Preliminary data demonstrate that 3 days of HDT (without raising the head on a pillow which immediately drains the brain circulation and lowers ICP) leads to thickening of the choroidal area, which we hypothesize may be the precursor to Spaceflight Associated Neuro-ocular Syndrome (SANS). Acute application of 20 mmHg LBNP lowers ICP values towards the upright posture, and daily use of 20 mmHg LBNP for 8 hours while awake prevents the choroidal expansion. Outcome variables for this aim will include choroidal volume, hemodynamic monitoring during sleep, systemic and cerebral hemodynamics using comprehensive ultrasound of the jugular veins, carotid and vertebral arteries, transcranial Doppler, non-invasive cardiac output, and blood and plasma volume. Invasive central venous pressure (PICC line) and IJVP also will be measured for comparison with the non-invasive IJVP techniques used at JSC.
Rationale for HRP Directed Research:	Insufficient time for solicitation to meet accelerated schedule of work to understand etiology and develop targeted countermeasures for Spaceflight Associated Neuro-ocular Syndrome.
Research Impact/Earth Benefits:	

Task Progress:	In this reporting period, we have procured all hardware and consumables, acquired all hardware from co-investigators, developed and prepared operator log sheets and study protocols, obtained NASA Institutional Review Board (IRB) approval, and held a study kick-off meeting with external co-investigators. In addition, we have completed hardware loan documentation to enable testing in Dr. Ben Levine's lab for the venous congestion countermeasure study (VCCM). To date, we have optimized all hardware for use in the study and trained multiple operators on use of the hardware. In addition, we have developed a lower body negative pressure system for the study within the Cardiovascular and Vision Laboratory, including repairing the LBNP chamber and developing an improved LBNP waist seal variable vacuum system. Furthermore, we have modified and tested our ultrasound data analysis program to accommodate the VCCM study design and requisite study variables and have modified our ultrasound data capture system to accommodate increased number of data collection sessions. This project requires use of a specialized OCT scan pattern and therefore we have verified that the OCT data is compatible with our analysis program and are in the process of establishing a subcontract with external investigators to conduct OCT-Angiography analysis. We have obtained NASA IRB approval and completed the test readiness review and to date, three dry run experiments have been performed. We are actively recruiting subjects and data collection for the official study is planned to start May 2019. NOTE (July 2020): Principal Investigator (PI) Dr. Michael Stenger moved to Research Operations and Integration element in February 2020; project taken over at that time by Dr. Karina Marshall-Goebel. For subsequent reporting, see project with the same title with Dr. Marshall-Goebel as the PI.
Bibliography Type:	Description: (Last Updated: 11/19/2025)